(a) Stochastic simulation model

Nine inputs θ (uniform)
Car ownership, car preference
Unemployment, free-time rate
Departure variation
Walking limit, bus headway
Random traffic, peak share

→

SUMO 1.28 chain, one seed ω
activitygen → duarouter → sumo
6×6 signalized grid, 120 edges
06:30–10:30, static routes
Run cache: hash(fingerprint, θ, ω)
Any failed run stops the pipeline

→

Outputs Y(θ, ω)
All-vehicle delay (mean SUMO time loss)
General-traffic delay (buses excluded)
Vehicle-kilometers traveled (VKT)
Diagnostics: trips by vehicle class, hourly arrivals, unfinished vehicles, teleports

(b) Pre-specified partitions (disjoint settings and seed blocks)

Development
512 scrambled Sobol
1 seed each
512 runs

Calibration
200 uniform settings
1 seed each
200 runs

Test (untouched)
200 uniform settings
5 seeds each
1,000 runs

Analysis protocol
Sizes, seeds, models and decision rules committed before the full run; models frozen and hashed

(c) Every surrogate claim is checked in SUMO

Surrogates
Fitted on development data only; fixed hyperparameters

Ridge
Gaussian process
Random forest
LightGBM
CatBoost
XGBoost
Ensemble (Cat + XGB)

Intervals: CQR, split conformal, GP, GP-CP (calibration set only)

Surrogate claim	Simulator check	
Accurate on new settings	Frozen models on the untouched test set; seed-noise R^2 ceiling and RMSE floor	1,240 runs
90% intervals cover new runs	Coverage per run, per setting and per congestion tertile on the test set	
Sensitivity ranking (Sobol, SHAP, ALE)	Direct-SUMO Sobol indices with seed-noise correction; 50 surrogate refits	5,632 runs
Bus headway reduces delay	8×8 factorial, 10 common seeds, two backgrounds, outcomes by vehicle class	1,280 runs
Intervention cuts delay 15%	Matched seeds: baseline, proposal and random change of equal size	600 runs
Calibration recovers θ	Excluded targets, fresh seeds, outputs outside the objective, equifinality	300 runs
Model applies to a new grid	8×8 grid: frozen transfer and four adaptation methods on fresh tests	428 runs

Total 10,216 SUMO runs; every number in the article is computed from saved result tables.